

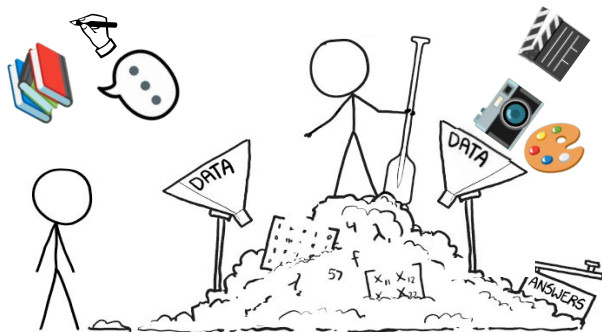
An introduction to (multicultural) vision and language models

Emanuele Bugliarelli
Research Scientist

3rd Cardiff NLP Summer Workshop
e-bug.github.io/assets/pdf/vlm_cardiffnlp24.pdf

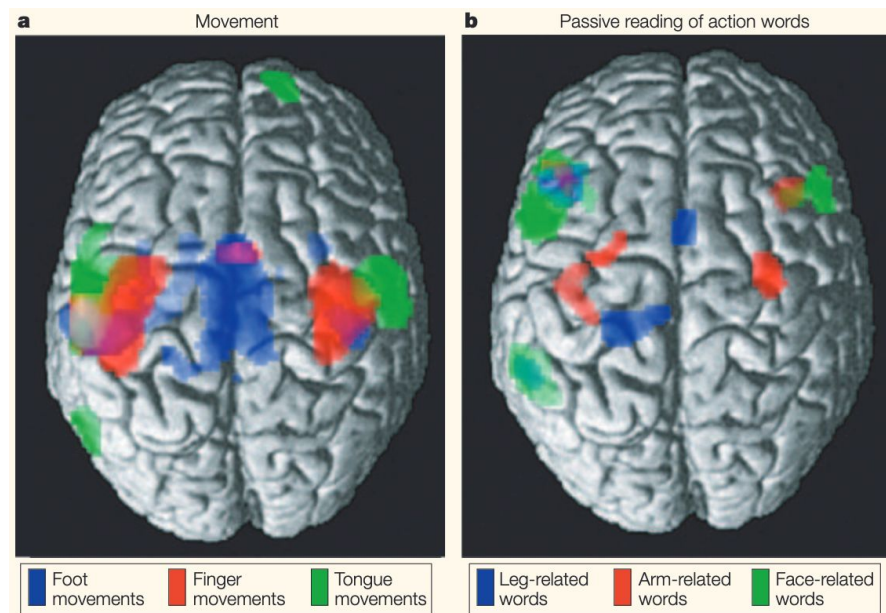
“ Multimodal Models: Working Definition

Multimodal models jointly process information from two or more modalities, e.g. images and text, speech and video, etc.



Why Multimodality?

- Humans ground conceptual knowledge in modality-processing systems in the brain
- Evidence that grounding activates similar brain regions for different input modalities

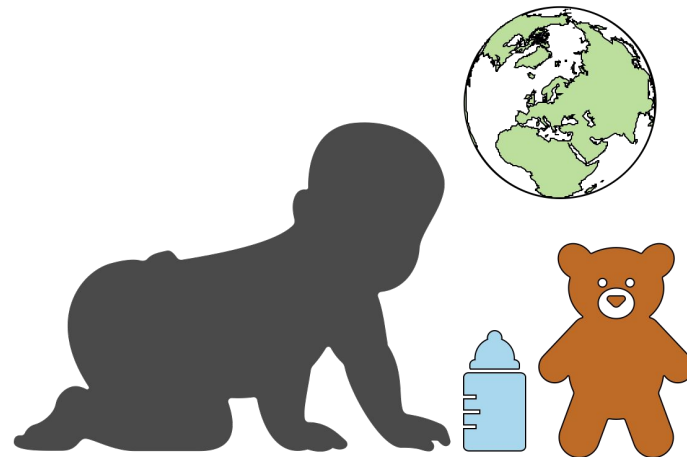


Humans vs. AI Learning



Today's AI paradigm

Learning from a single form of data (i.e., modality)



Humans

Learning from multiple modalities

Contents

Multimodal Representation Learning	1
Multimodal Data	2
Cultural Considerations	3
Conclusions	4

Multimodal Representation Learning

Enhancing language representations with perceptual
information



Representing Visual Data

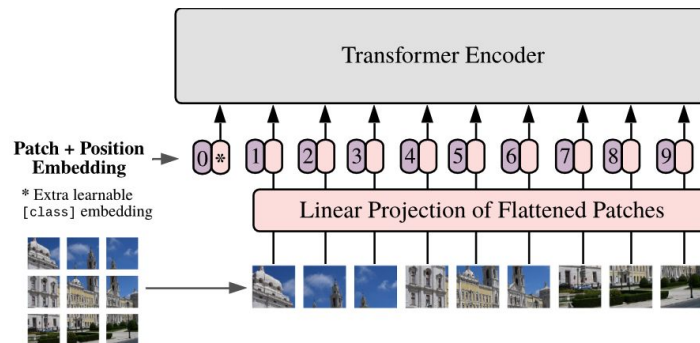
Representing Visual Data: Vision Transformers

Good news! You are already almost an expert in how the Vision Transformer works

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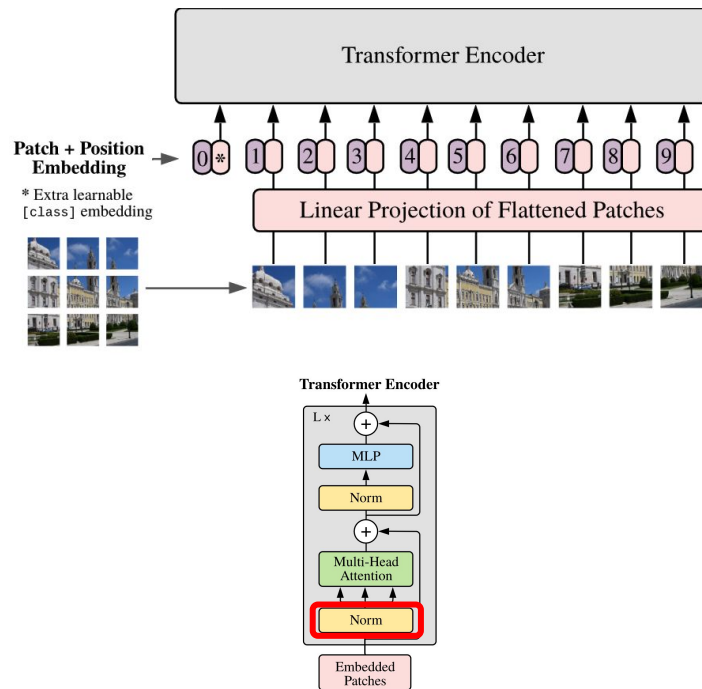
- Split image into K patches
- Embed each patch
- Add position information



Representing Visual Data: Vision Transformers

Good news! You are already almost an expert in how the Vision Transformer works

- Split image into K patches
- Embed each patch
- Add position information
- Encode using Transformer blocks that include an extra **pre-norm layer** for training stability

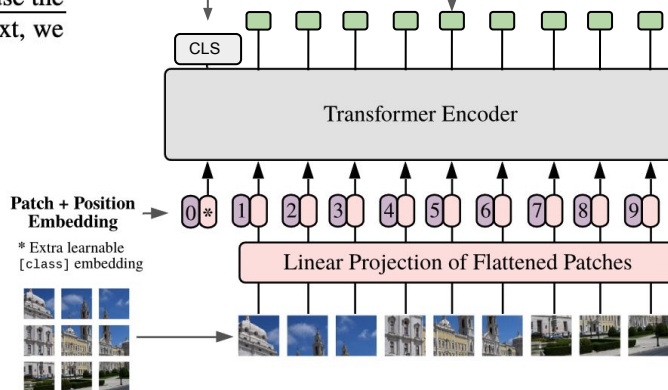


Representing Visual Data: Vision Transformers

- Extract **pooled features** or **patch-level features**

To extract visual information from an image x^i , we use the visual encoder of a pre-trained CLIP [29] model. Next, we

For the CLIP encoders, we extract the feature grid before the pooling layers, resulting in an $N \times N$ grid, where $N = 7, 7, 12$ for the ViT-B/32, RN50x4 and RN50x16 variants of CLIP respectively.



Main Multimodal Approaches

Cross encoding

Vision+Language Model

The red
horse



Dual encoding

Language

The red
horse

Vision



Visual Prefix

Language Model

Vision

The red
horse



Cross-encoding Models

- A single, multimodal model
- Emerged as a key modelling approach in 2019 with a flurry of approaches to creating visually grounded BERT models

Cross encoding

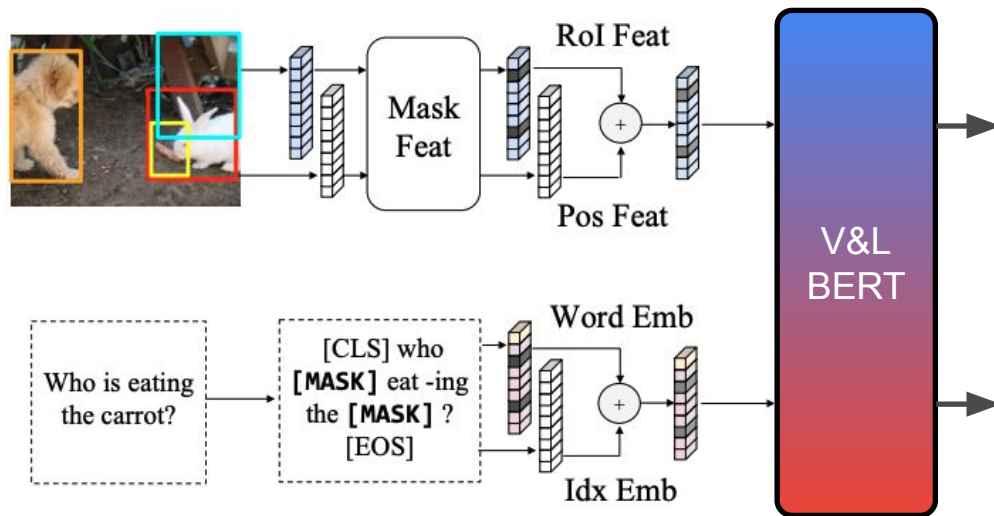
V+L Transformer

The red
horse



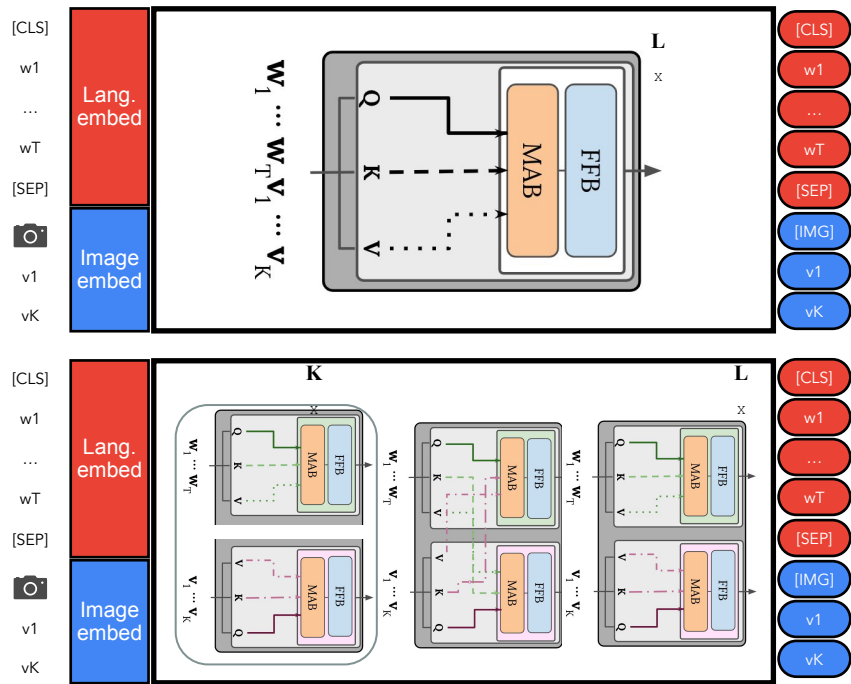
Cross-encoding Models: Input Embeddings

- *Image*
 - Faster R-CNN
 - Raw pixels + ViT
- *Language*
 - BERT tokens



Cross-encoding Models: Single- & Dual-Stream Architectures

- Single-stream
 - Concatenate inputs
- Dual-stream
 - Process modalities separately
 - Intra-modal
 - Inter-modal



Cross-encoding Models: Typical Losses

- Masked Language Modelling
 - Same as BERT et al.

$$\mathcal{L}_{\text{MLM}}(\theta) = -\mathbb{E}_{(\mathbf{w}, \mathbf{v}) \sim D} \log P_{\theta}(\mathbf{w}_{\mathbf{m}} | \mathbf{w}_{\setminus \mathbf{m}}, \mathbf{v})$$

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- Masked Region Classification
 - Mean Squared Error Regression over the 2048D feature vector; or
 - Predict the probability distribution over the 1600 Faster R-CNN classes

$$\mathcal{L}_{\text{MRM}}(\theta) = \mathbb{E}_{(\mathbf{w}, \mathbf{v}) \sim D} f_{\theta}(\mathbf{v}_{\mathbf{m}} | \mathbf{v}_{\setminus \mathbf{m}}, \mathbf{w})$$

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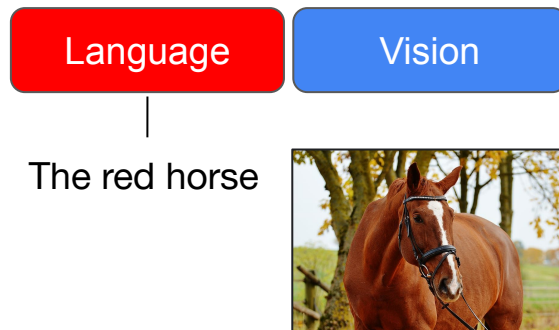
- Image-Text Matching

- 50% chance of randomly sampling a mis-matched sentence
- Predict with a binary classifier (aka Next Sentence Prediction)

$$\mathcal{L}_{\text{ITM}}(\theta) = -\mathbb{E}_{(\mathbf{w}, \mathbf{v}) \sim D} [y \log s_{\theta}(\mathbf{w}, \mathbf{v}) + (1 - y) \log(1 - s_{\theta}(\mathbf{w}, \mathbf{v}))]$$

Dual-encoding Models

- Emerged as a sample-efficient alternative to cross-encoding
- The backbone consists of two separate components:
 - language encoder
 - visual encoder

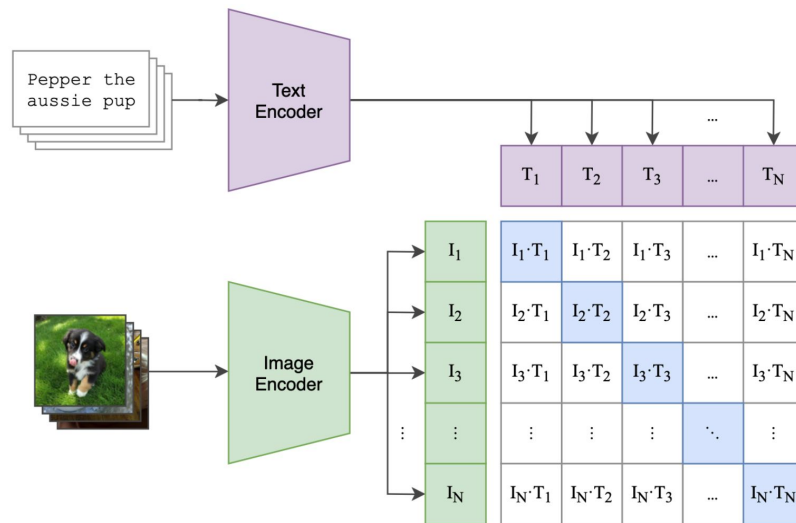


Dual-encoding Models: CLIP

- Transformer Text Encoder
- ViT or ResNet Visual Encoder
- Maximize the similarity of the embeddings of paired examples (t, i):

$$-\frac{1}{2|\mathcal{B}|} \sum_{i=1}^{|\mathcal{B}|} \left(\overbrace{\log \frac{e^{\mathbf{x}_i \cdot \mathbf{y}_i}}{\sum_{j=1}^{|\mathcal{B}|} e^{\mathbf{x}_i \cdot \mathbf{y}_j}}}^{\text{image} \rightarrow \text{text softmax}} + \overbrace{\log \frac{e^{\mathbf{x}_i \cdot \mathbf{y}_i}}{\sum_{j=1}^{|\mathcal{B}|} e^{\mathbf{x}_j \cdot \mathbf{y}_i}}}^{\text{text} \rightarrow \text{image softmax}} \right)$$

where $\mathbf{x}_i = \frac{f(I_i)}{\|f(I_i)\|_2}$ and $\mathbf{y}_i = \frac{g(T_i)}{\|g(T_i)\|_2}$.

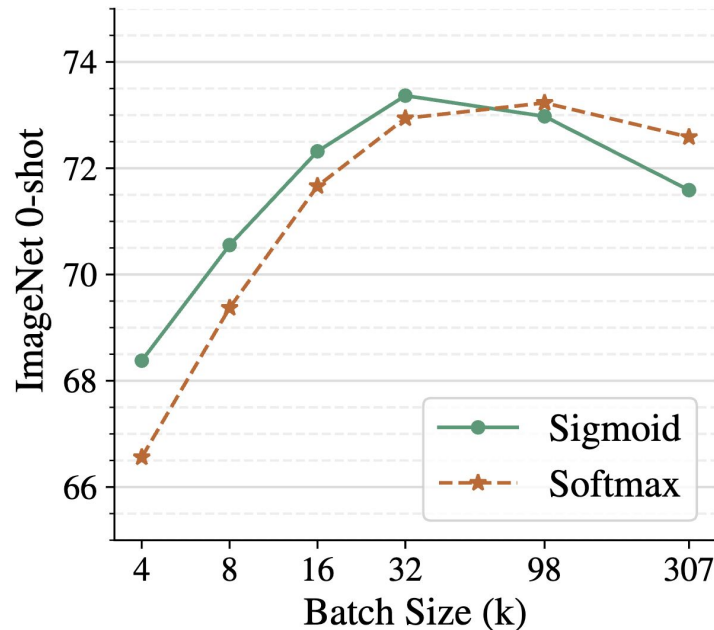


Dual-encoding Models: SigLIP

- From softmax (global normalization factors) to sigmoid (binary classification):

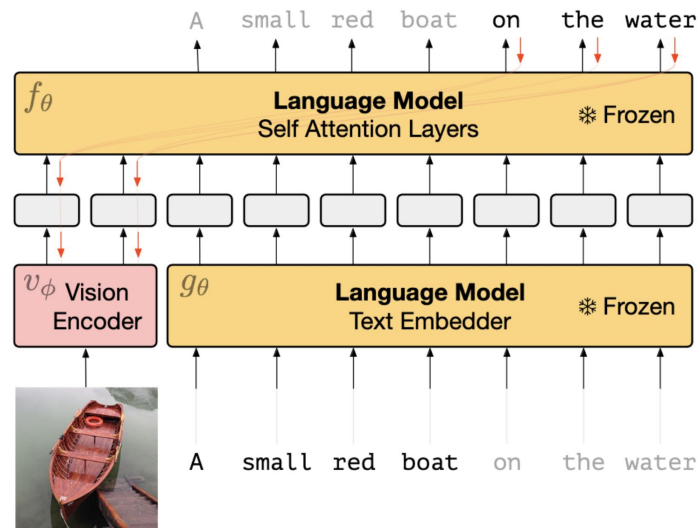
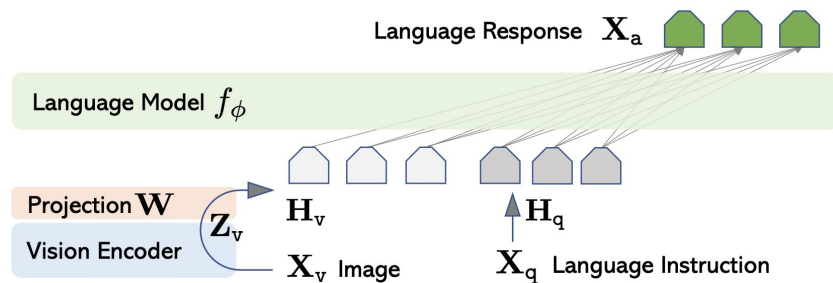
$$-\frac{1}{|\mathcal{B}|} \sum_{i=1}^{|\mathcal{B}|} \sum_{j=1}^{|\mathcal{B}|} \underbrace{\log \frac{1}{1 + e^{z_{ij}(-t\mathbf{x}_i \cdot \mathbf{y}_j + b)}}}_{\mathcal{L}_{ij}}$$

where z_{ij} is the label for a given image and text input, which equals 1 if they are paired and -1 otherwise.



Visual Prefix Models

- Exploit the representations learned during large-scale modality-specific pretraining

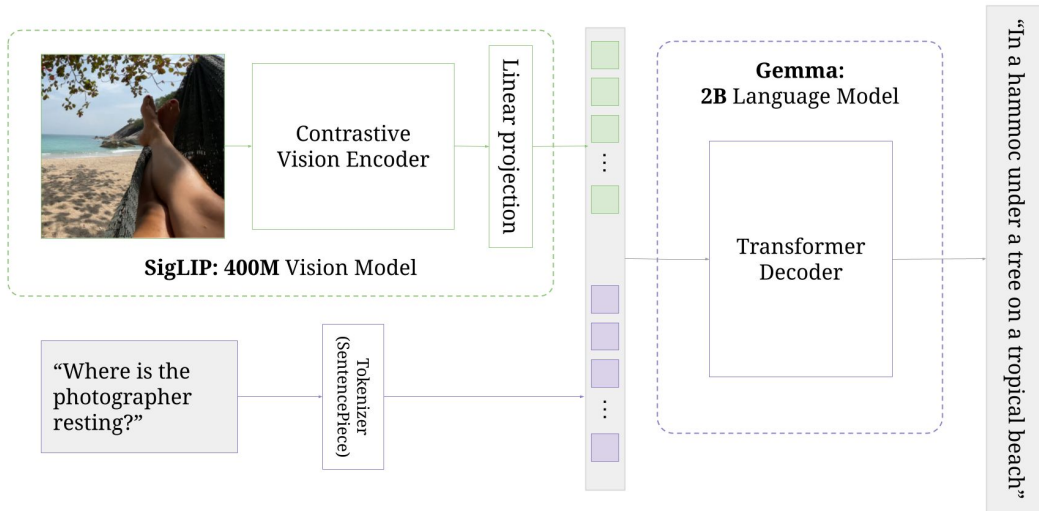


Visual Prefix Models: PaliGemma

A family of lightweight open vision-language models, designed for class-leading fine-tune performance on a wide range of vision-language tasks

Training

1. Unimodal pretrain
2. Multimodal end-to-end train
3. Multimodal high-res train
4. Task fine-tuning



Visual Prefix Models: PaliGemma for the Community

PaliGemma open models make vision-language models extremely accessible to developers, researchers and practitioners

- 3B parameters
- Free for everyone
- 20 min to fine-tune in a Google Colab with a single free T4 GPU



Fine-tuning Colab: <https://t.co/7XCNrcKKWf>

hf.co/spaces/big-vision/paligemma

Visual Prefix Models: PaliGemma Results

Benchmark	Metric (split)	Task	Results	Previous Best Open Models
POPE	Accuracy (random / popular / adversarial)	Object Hallucination	88.4	87.2
VQAv2	Accuracy (Test server - std)	Visual Question Answering	85.6	83.7
ScienceQA	Accuracy (test)	Scientific Question Answering	95.9	94.7
RSVQA-HR	Mean Accuracy (test / test2)	Remote Sense Question Answering	92.8 90.5	92.1 89.8
MSR-VTT	Accuracy (test)	Video Question Answering	50.1	48.0

Visual Prefix Models: PaliGemma Capabilities

What does the image show?



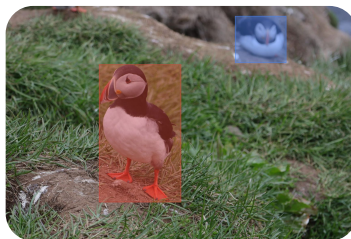
A **small house** sits nestled amidst a **serene landscape**, its **red exterior** contrasting with the vibrant green grass and towering mountains. The house exudes a sense of ...

Read the sign for me



notice: motorcyclists please do not hold your boarding pass in your mouth

Detect puffin in the back; puffin in front



puffin in the back puffin in front



Q: Which breed is the fox?
A: arctic

Q: Is the fur short or long?
A: long

Q: Is the tail fluffy or sleek?
A: fluffy

Multimodal Data



What kind of multimodal data exists?

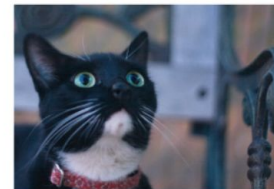
Two Types of Datasets

- *General-purpose*: visual data with descriptive annotations
 - Conceptual Captions
 - LAION-2/5B
 - Speech-COCO
- *Task-specific*: visual data with e.g. classification labels
 - Image / Video Captioning
 - Visual Question Answering
 - Visually Grounded Reasoning



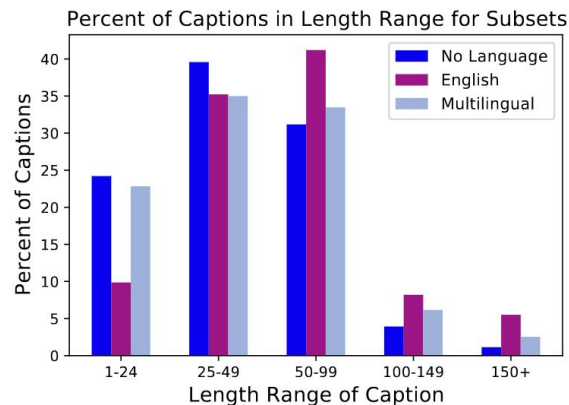
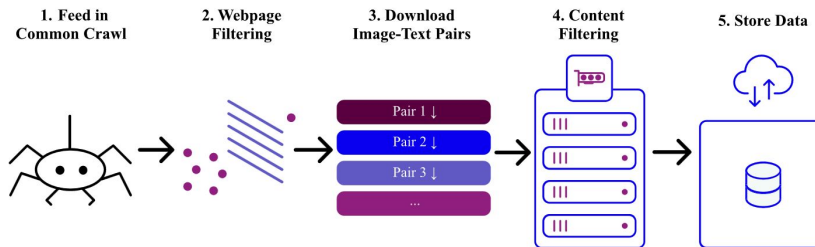
Blue Beach
Umbrellas, Point
Of Rocks, Crescent
Beach, Siesta Key -
Spiral Notebook

What color is the cat's leash?
purple red



LAION

- Used for pretraining
- Image and multilingual raw captions harvested from Common Crawl
- Data behind Stable Diffusion and OpenCLIP
- 5B variant removed due to illegal material



COCO

- Used both a general-purpose and task-specific dataset
- ~300k images covering 80 common objects in context with multiple human-authored captions
- Object segmentation data too!

some sheep walking in the middle of a road
a herd of sheep with green markings walking down the road
a herd of sheep walking down a street next to a lush green grass covered hillside.
sheared sheep on roadway taken from vehicle, with green hillside in background.
a flock of freshly sheared sheep in the road.



VQAv2

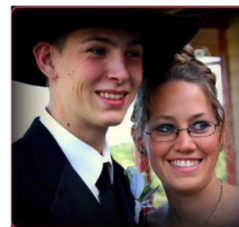
- Answer questions about images
- Task with multimodal inputs:
 - Image
 - Question
- 1.1M image-question pairs with balanced distribution of answers
- Commonly tackled as classification but increasing efforts as NLG

Who is wearing glasses?

man



woman

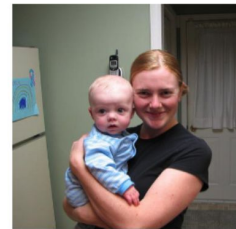


Where is the child sitting?

fridge



arms



NLVR2

- Binary classification task that requires jointly reasoning over a pair of images and a sentence
- Human-created hard negatives
- 107K examples in total



The left image contains twice the number of dogs as the right image, and at least two dogs in total are standing.

Many more tasks and datasets

- Visual Storytelling, e.g. VIST
- Grounded Referring Expression, e.g. Flickr30K Entities, Visual Genome
- Visual Entailment, e.g. SNLI-VE
- Vision & Language Navigation, e.g. RxR
- Visual Common Sense Reasoning, e.g. VCR
- Text-to-Image Generation, e.g. DALLEval
- Abstract reasoning, e.g. KiloGram, CRAFT
- Sign Language Processing, e.g. How2Sign
- and more and more and more and more

Rethinking Vision and Language Data

- *Languages*

- Mostly in English
- Or some Indo-European Languages



ENG: An **unusual** looking vehicle ...

NLD: Een mobiel **draaiorgel** ...

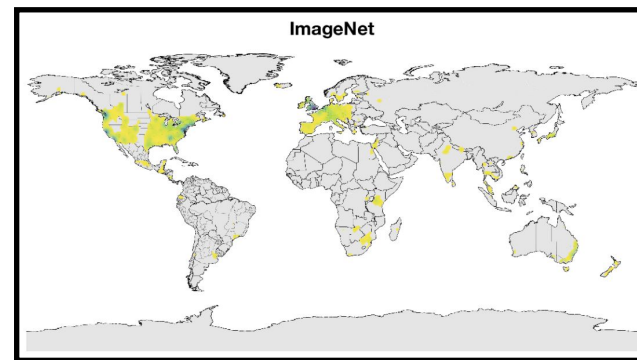
Example from [van Miltenburg+ 2017](#)

- *Image sources*

- Mostly from ImageNet or COCO
- Reflecting North American and European cultures

- *Implications for V&L models*

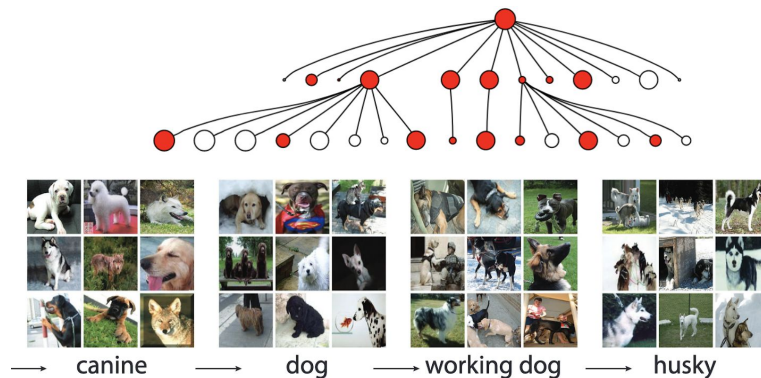
- Narrow linguistic/cultural domain
- No way to assess their real-world comprehension



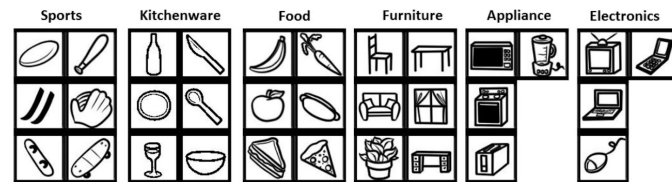
Density map of geographical distribution of images in ImageNet ([DeVries+, 2019](#))

Typical Vision and Language Data

- *ImageNet* (Deng et al. 2009)
 - Train visual encoders
 - Millions of labelled images
 - Derived from the WordNet hierarchy

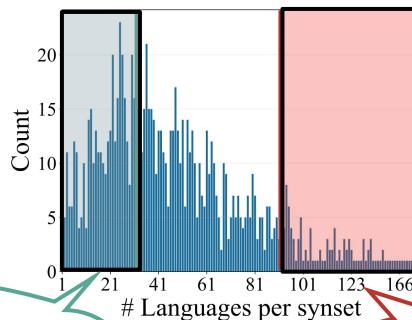
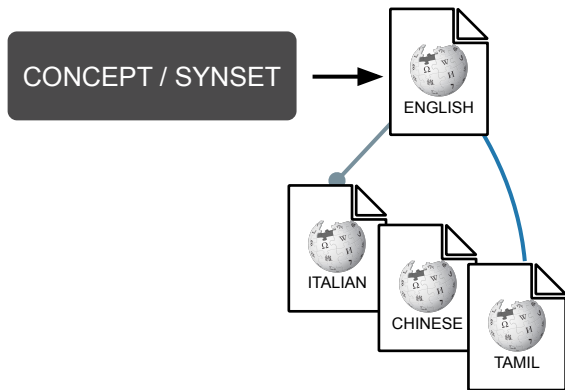


- *COCO* (Lin et al. 2014)
 - Train and evaluate multimodal models
 - 330K labelled images
 - 80 types of commonly occurring objects



Are ImageNet Concepts Cross-Lingual?

- The ImageNet, COCO and Visual Genome datasets use English WordNet concepts
- Idea: estimate cross-linguality using Wikipedia as a proxy



Most concepts in
<30 languages

Only a few "universal"
concepts

MaRVL: Multicultural Reasoning over Vision and Language



Representative of annotators' cultures



5 typologically diverse languages

Independent, culture-specific annotations



MaRVL-id Bola basket



MaRVL-sw Mpira wa kikapu



MaRVL-tr Basketbol



MaRVL-zh 篮球



MaRVL-ta
கூடைப்பந்தாட்டம்

Visual Reasoning Task (NLVR2; Suhr et al., ACL 2019)

- Datapoint: two images (v_1, v_2) paired with a sentence x
- Task: Predict whether x is a true description of the pair of images (v_1, v_2)



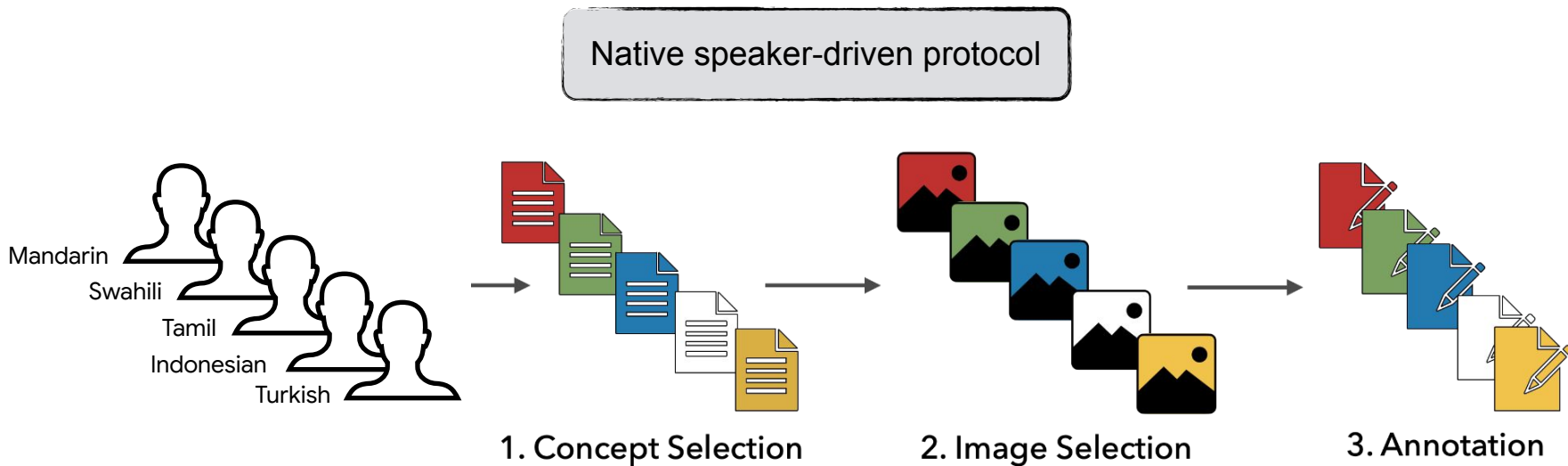
இரு படங்களில் ஒன்றில்
இரண்டிற்கும் மேற்பட்ட
மஞ்சள் சட்டை அணிந்த
வீரர்கள் காளையை அடக்கும்
பணியில் ஈடுபட்டிருப்பதை
காணமுடி.

True

x

y

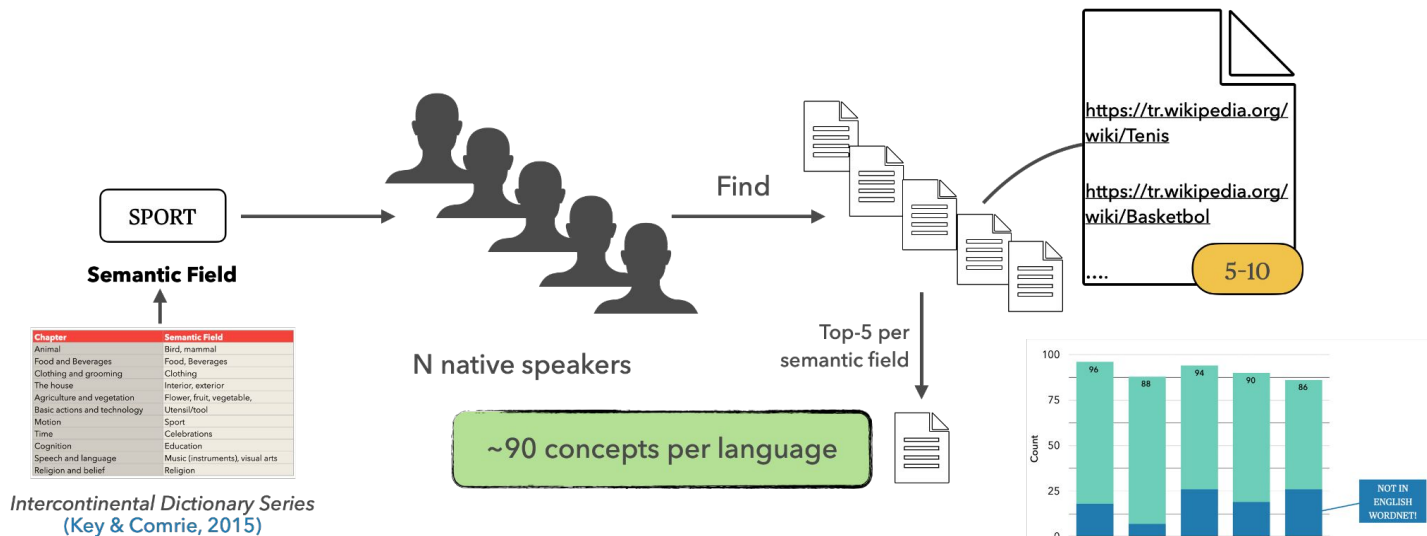
Collecting MaRVL data



Step 1. Language-Specific Concept Selection

Collected by native speakers

- Concepts are commonly seen or representative in their culture
- Concepts are ideally, physical and concrete



Step 2. Image Collection

Collected by native speakers

- Representative of the language population
- NLVR2 requirements:
 - Contains more than one instance of a concept
 - Shows an instance of the concept interacting with other objects
 - Shows an instance of the concept performing an activity
 - Displays a set of diverse objects or features



MaRVL-zh 花椰菜 (Cauliflower)



MaRVL-ta கெட்டி (Buttermilk)



MaRVL-sw Jembe (Shovel)



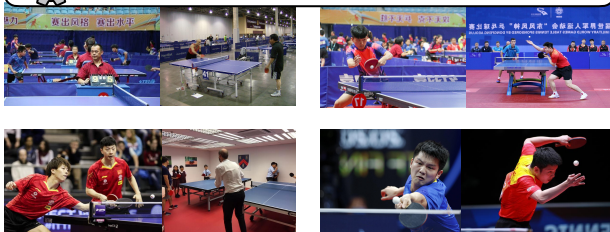
MaRVL-tr Rakı (Raki)

Step 3. Language Annotation

Collected by native speakers



MATCH 4 PAIRS AT RANDOM



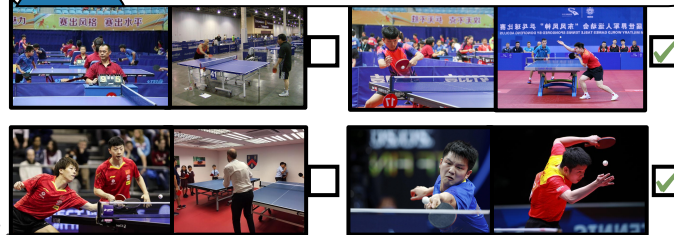
VALIDATE ANNOTATIONS



右图中的人在发球, 左图中的人在接球。



WRITE CAPTION TRUE ONLY FOR 2 PAIRS



右图中的人在发球, 左图中的人在接球。



FINAL VALIDATION



右图中的人在发球, 左图中的人在接球。

(The man in the right image is serving a ball while the man in the left image is returning a ball.)

Fleiss' kappa: 93%

Dataset Examples

MaRVL-tr Kanun (çalgı)



Görsellerden birinde dizlerinde kanun bulunan birden çok insan var

(In one of the images, there are multiple people with qanuns on their knees)

Label: True

MaRVL-ta வை (Vada)

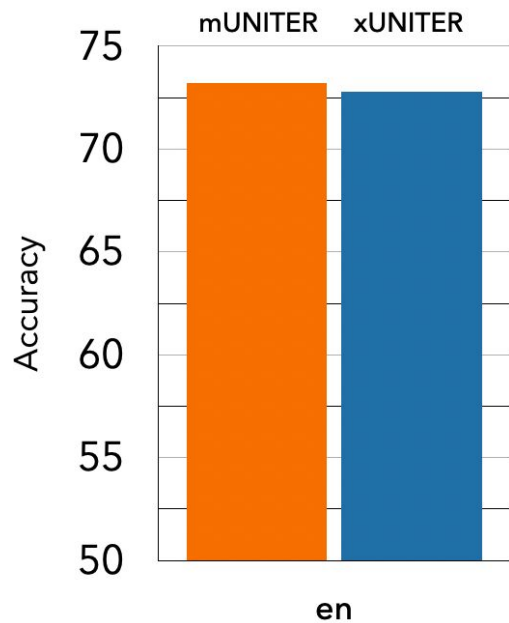


இரண்டு படங்களிலும் நிறைய மசால் வடைகள் உள்

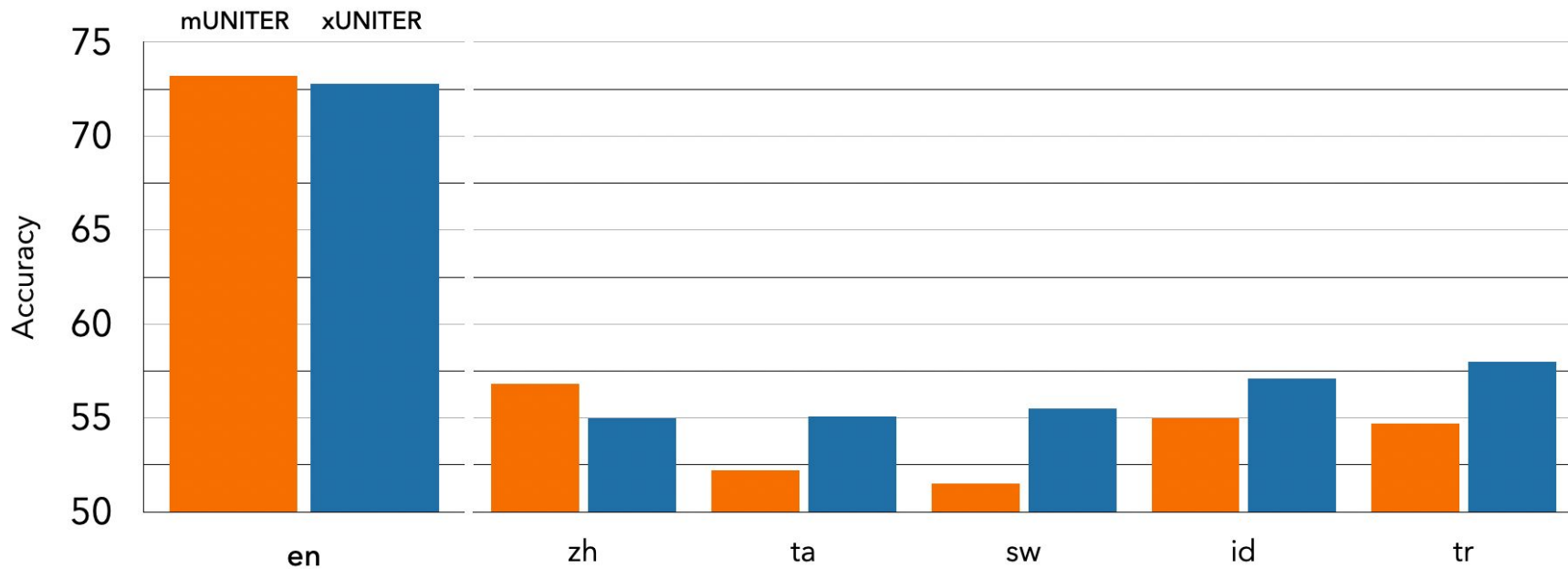
(Both images contain a lot of masala vadas)

Label: False

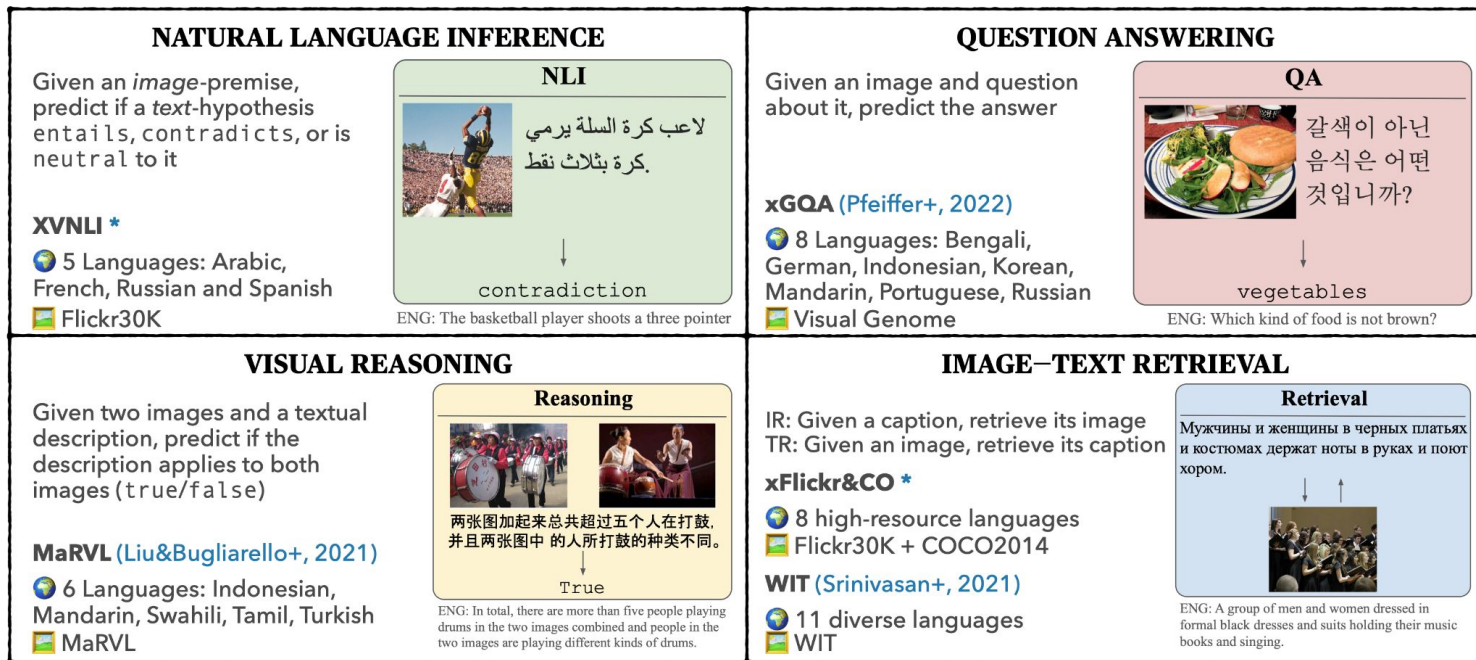
MaRVL Zero-shot Results



MaRVL Zero-shot Results

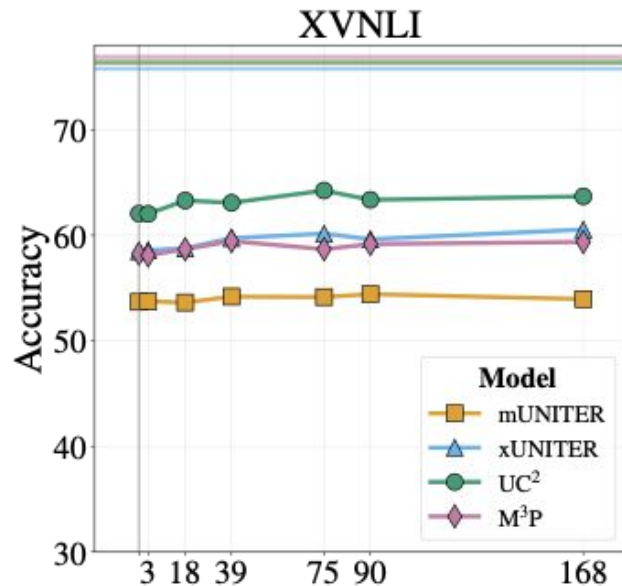


IGLUE: A Multilingual Multimodal Multi-task Benchmark in 20 Languages



IGLUE: Few-shot Learning Results

- Performance on XVNLI as a function of the number of multilingual samples
- Horizontal lines are Eng performance
- Few-shot multilingual learning is not trivial in multimodal setups

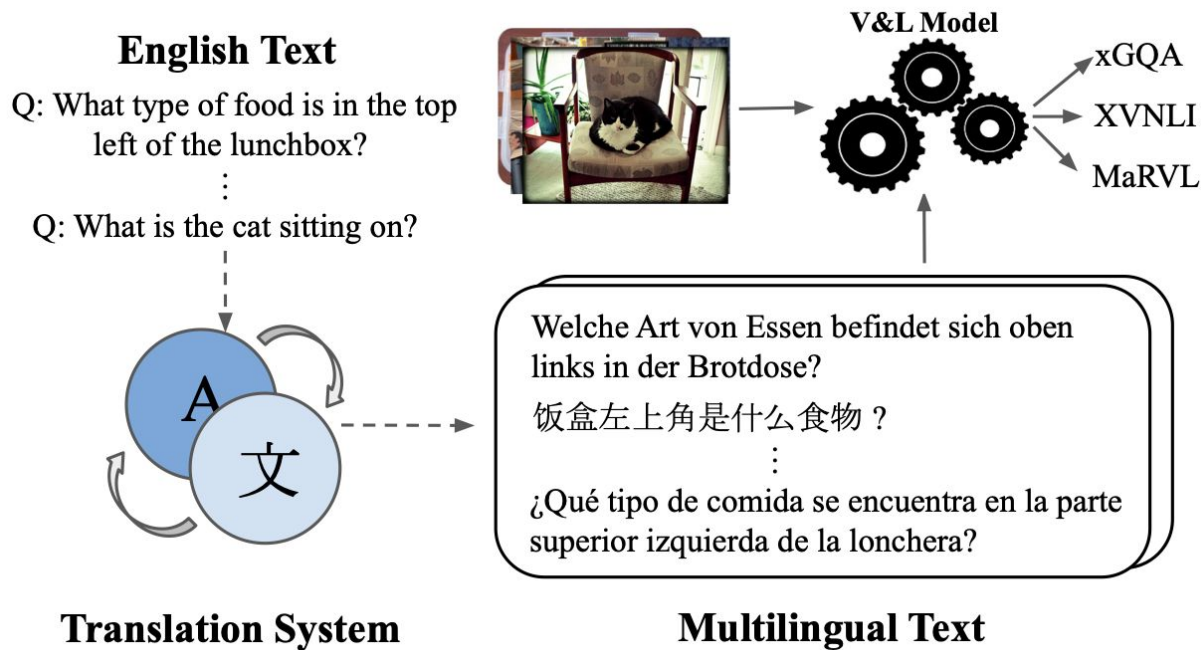


Closing the English Gap

Enhancing language representations with perceptual information

A large, bold, blue number '3' is positioned on the right side of the slide. It is a simple, sans-serif font with a thick stroke.

TD-MML: Translated Data Multilingual Multimodal Learning



TD-MML: Translation Methodology

- Translate 2.77M sentences from the English Conceptual Captions dataset into the 20 IGLUE languages using M2M-100_{LARGE}
- Filter out potentially bad data in the 55.4M translated sentences:
 - *Repeated tokens*: type-to-token ratio of target sentence
 - *Copied tokens*: sBLEU against source sentence

damask seamless floral pattern, ornament → *Mifano ya Mifano ya Mifano ya Mifano ya Mifano ya Mifano ya Mifano ya Mifano ya Mifano (SWA)*

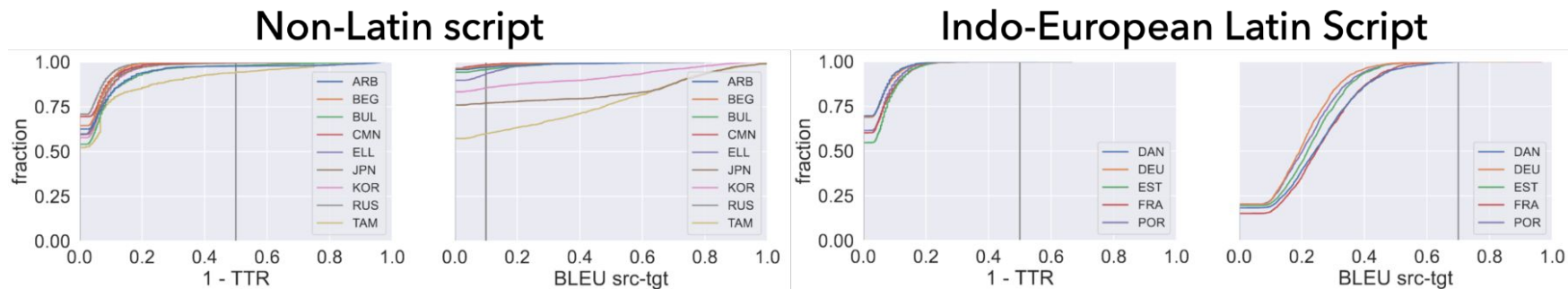
BAD TRR

plaid, over garment, outfit idea cute fall outfit idea → *方格, over garment, cute fall (CMN)*

HIGH
SBLEU

TD-MML: Filtering Thresholds and Resulting Corpus

- Filtering threshold depends on script and language family



- Many copied tokens in non-Latin script languages
- Final pretraining corpus has 52M sentences
 - Japanese, Korean and Tamil are the most affected by our filtering process

TD-MML: Zero-shot IGLUE Results

- Clear improvement compared to our baselines

Model	NLI	QA	Reasoning	Retrieval			
	XVNLI	xGQA	MaRVL	xFlickr&CO		WIT	
				IR	TR	IR	TR
mUNITER	53.69	9.97	53.72	8.06	8.86	9.16	10.48
xUNITER	58.48	21.72	54.59	14.04	13.51	8.72	9.81
UC ²	62.05	29.35	57.28	20.31	17.89	7.83	9.09
M ³ P	58.25	28.17	56.00	12.91	11.90	8.12	9.98
TD-MML	64.84	35.95	59.67	21.30	26.35	9.76	10.35

No Filter: Cultural and Socioeconomic Diversity in Contrastive VLMs

- Most models pretrain on English-only data
- Better image features from global data based on geo-localization tasks
- NB: We focus on visual world knowledge, disentangling from multilinguality



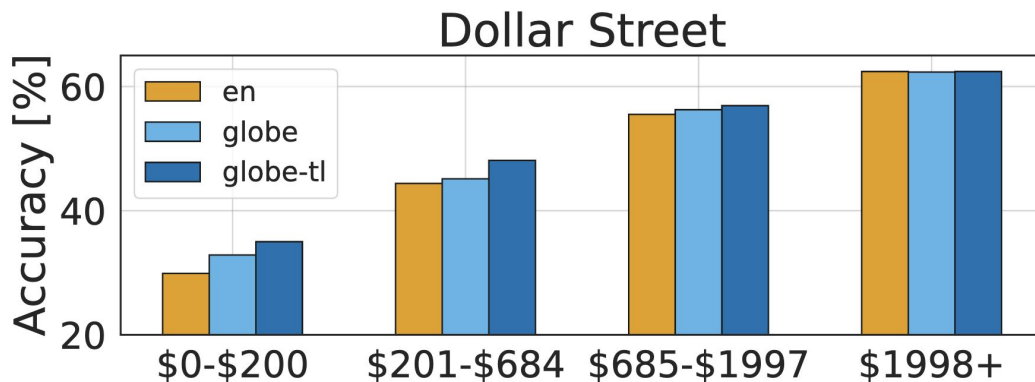
(a) **en:** CN Tower
(Canada)
globe: Milad Tower
(Iran)



(b) **en:** Notre-Dame Basilica of Montreal (Canada)
globe: Catedral da Sé de São Paulo (Brazil)

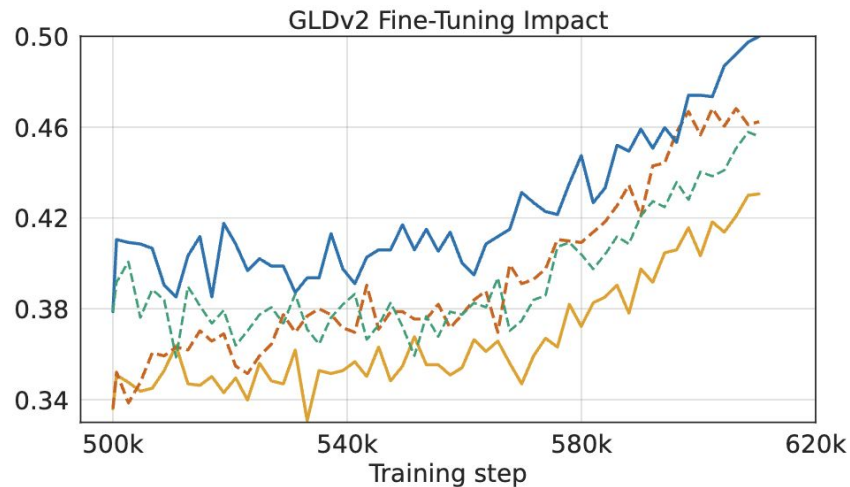
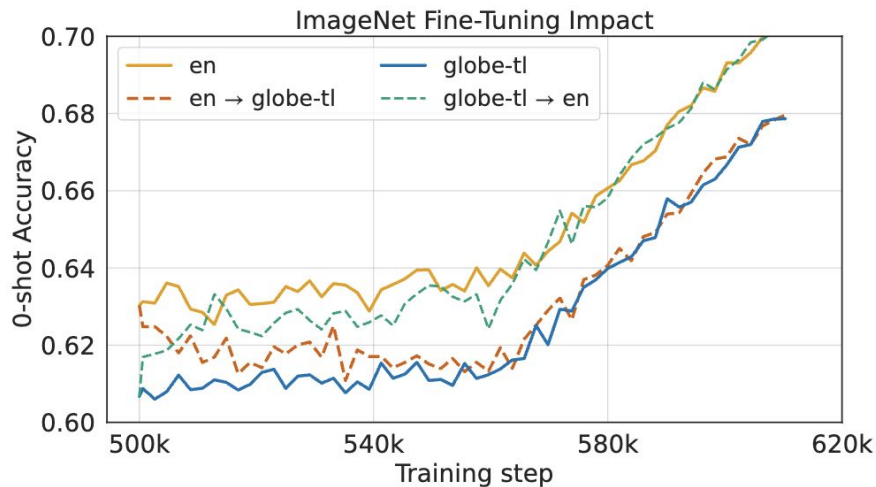
No Filter: English-only data exacerbates socioeconomic disparities

- Three training setups:
 - *en*: English-only data
 - *globe*: raw, multilingual data
 - *globe-tl*: globe data but with non-English text machine-translated to English



No Filter: Global models can adapt to Western data, but not vice versa

- Fine-tuning *globe-tl* on en data quickly catches up on Western benchmarks
- Fine-tuning *en* on globe-tl data does not close the gap in culturally diverse benchmarks



Conclusions

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Recap & Future Work

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 - Cross-encoders vs. Dual-encoders vs. Visual prefix
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- Shift in focus to zero-shot instruction-based models
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Thank you!

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